

Optimisation for Machine Learning

» Admin

Dr. Mouli Chakraborty and Prof. Douglas Leith

email : mchakrab@tcd.ie

- * Module material: on Blackboard
- * Recommended text book: Algorithms for Optimization, Kochenderfer & Wheeler, 2019
<https://algorithmsbook.com/optimization/files/optimization.pdf>
- * Online resources are often quite mathsy, but we'll be more hands-on.
- * 40% marks for weekly assignments, 60% for individual project/assignment
- * Prerequisites: python programming, a first course in machine learning e.g. CS7CS4

python

» Honour Code

Keep in mind (Please) !!!

- * Its ok (good in fact) to discuss the weekly assignments with other students, but you *must*:
 - (i) Write the answers yourself in your own words
 - (ii) Write your own code, and don't share your code with others
 - (iii) You must explain/justify how you obtained your answer. So if you write code to carry out a calculation you need to discuss/explain what that code does, and if you present numerical results you need to discuss their interpretation. Saying "see code", simply giving numbers or plots without comment → low mark.
- * Individual project must be entirely your own work (no collaboration/discussion).
- * All submissions will be checked by anti-plagiarism software and each week a random sample will also be checked manually.
- * *Zero tolerance for plagiarism - referral to college authorities for disciplinary action.*

» Optimisation & Machine Learning

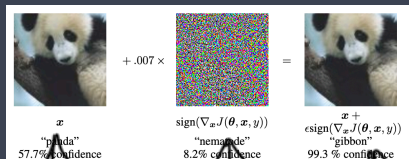
This is the data science strand, so why study optimisation? Recall usual supervised ML setup:

- * Labelled training data (x_i, y_i) $i = 1, \dots, m$, model $\hat{y} = h_{\theta}(x)$ with unknown parameters θ
- * Learning/Training: choose model parameters that minimise a cost function e.g. $\sum_{i=1}^n (y_i - h_{\theta}(x_i))^2$
- * *Minimising a cost function $\rightarrow$ optimisation*
- * Some examples of ML optimisation: training linear regression, logistic regression, neural net models
- * Also hyperparameter selection $\rightarrow$ optimisation

» Optimisation & Machine Learning

Attacks against ML models e.g.

- * Linear model output/prediction: $\hat{y} = \theta^T x$.
- * Suppose attacker can perturb input x by adding vector η to it, model output/prediction becomes $\hat{y} = \theta^T (x + \eta)$.
- * How can attacker choose η so as to maximally disrupt model output/prediction?
- * Define reward function that is large when model output/prediction $\theta^T (x_i + \eta)$ is a poor match to the training data output y_i . Then select η to *maximise* this reward, perhaps subject to *constraint* that η is sparse or has small magnitude (so looks like “noise”) → *optimisation*



<https://arxiv.org/pdf/1412.6572.pdf>

input without noise noise output with noise

» Optimisation & Machine Learning

Another example: multi-arm bandit/online learning

- * In A/B testing want to decide which of two options is best e.g. two UI designs. Usual approach: present option A to some people, option B to other people, measure performance, make decision about whether A or B is better.
- * Multi-arm bandit: repeatedly select one of N arms ($N = 2$ in case A/B testing), observe noisy loss. Want to minimise aggregate loss over time.
- * Pull arm $i_t \in \{1, 2, \dots, N\}$ at turn t , observe loss L_{i_t} . Want to select sequence of arm pulls $i_1, i_2, \dots, i_n$ such that $\sum_{t=1}^n L_{i_t}$ is minimised $\rightarrow$ *optimisation*



Shortest path

*Example
Graph
nodes*

» Optimisation Notation

- * Unconstrained optimisation:

$$\min_x f(x)$$



- * x is a vector $[x_1, x_2, \dots, x_n]$, sometimes called the decision variables
- * $f(\cdot)$ is function to be minimised. It maps from vector x to a real (scalar) value e.g. $f(x) = x_1^2 + x_2^2$ with $x = [x_1, x_2]$
- * Its usual to refer to $f(\cdot)$ as a cost function, the idea being that we want to minimise costs.
- * Canonical way to write an optimisation problem is as a minimisation. But suppose we want to maximise a function $g(\cdot)$. Write as

$$\max_x g(x)$$

- * When maximising its usual to refer to $g(\cdot)$ as a reward function, the idea being that we want to maximise rewards.

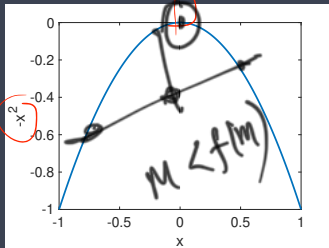
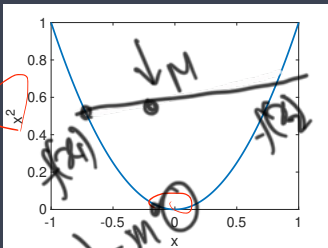
if conditions are given e.g.
 $x \leq 1$, or $x \geq 1$
constrained part

» Optimisation Notation

- * Can always change a maximisation into a minimisation by taking negative i.e.

$$\max_x g(x) = \min_x -g(x)$$

Why?



How to identify?

$M > f(x)$

$f(x) = M$
convex

$(x_2, f(x_2))$

$M < f(x)$
concave

(x, y)
 $f(x)$

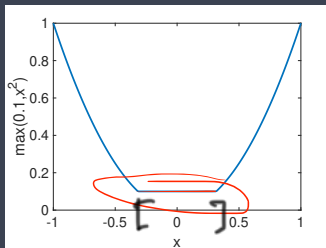
» Optimisation Notation

- * The set of values of x that minimize function $f(\cdot)$ is

$$\arg \min_x f(x)$$

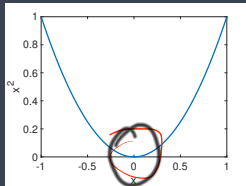
So if $f_{\min} = \min_x f(x)$ then $f(x^*) = f_{\min}$ for $x^* \in \arg \min_x f(x)$

- * Why a set? There can be more than one value that minimises a function e.g.

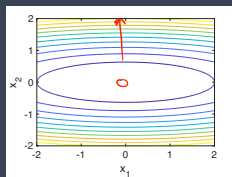
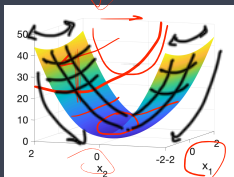


» Cost “Landscape” or “Surface”

- * When thinking about minimising a cost function $f(\cdot)$ its often helpful to plot it. E.g.



quadratic function of one variable



quadratic function of two variables



↑
← contour plot

- * For functions of 2 variables we have a 3D plot. Often hard to visualise (and draw!) so its common to use a contour plot
- * For function of > 2 variables we can't plot it, but sometimes use a 2D sketch to discuss.

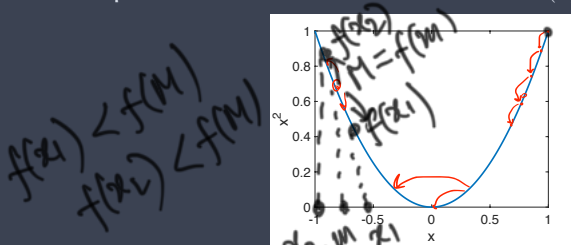
» To find minimum, move downhill

Example: suppose we want to find x that minimises $f(x) = x^2$.
Brute force search over possible values of x is inefficient, can we be smarter ?

- * Start with some x , e.g. $x = 1$.

- * Repeat:

Update x to new value which makes $f(x)$ smaller



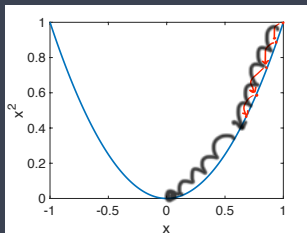
- * When curve is “bowl shaped” or convex then this must eventually find the minimum.

» To find minimum, move downhill

Example: suppose we want to find x that minimises $f(x) = x^2$.
Brute force search over possible values of x is inefficient, can we be smarter ?

- * Start with some x , e.g. $x = 1$.
- * Repeat:
 - Update x to new value which makes $f(x)$ smaller i.e. move “downhill”

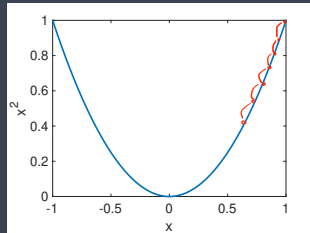
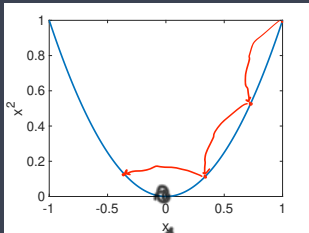
~~11~~



- * When curve is “bowl shaped” or convex then this must eventually find the minimum.

» To find minimum, move downhill

What size of downhill “step” to take?



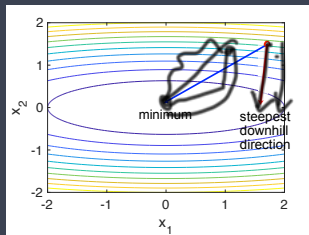
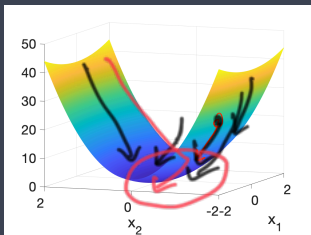
Too big: we jump right past the minimum, *Too small*: takes ages to find minimum.



» To find minimum, move downhill

How do we find a downhill direction? What is the “best” direction?

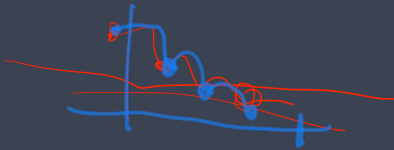
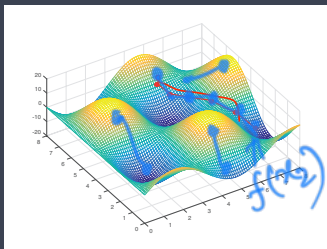
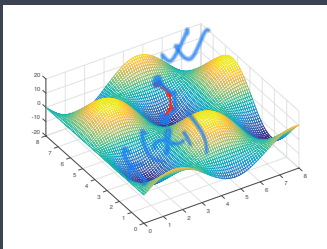
- * Its easy in our toy example, but often not so easy in practice.
E.g. suppose our cost function has a narrow valley



» To find minimum, move downhill

Global or local minimum?

- * When curve has several minima (think “mountain range” rather than single valley) then we can’t be sure which we will converge to.
- * Might not converge to the global minimum



» To find minimum, move downhill

Other important matters

- * When to stop?
- * How quickly can we find minimum?
- * How does shape/properties of function affect time taken to find minimum?

